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In [2]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

FILE = r"D:\Datascience-MachineLearning\Datasets\titanic.csv"

df = pd.read_csv(FILE)
df = df[["Survived", "Pclass", "Sex", "Age", "Fare", "SibSp", "Parch"]].copy()
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Sex"] = df["Sex"].map({"male": 0, "female": 1})
df = df.dropna()

X = df.drop(columns="Survived")
y = df["Survived"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
pred = knn.predict(X_test)

acc = accuracy_score(y_test, pred)
cm = confusion_matrix(y_test, pred)

print("KNN accuracy:", acc)
print("Confusion Matrix:")
print(cm)

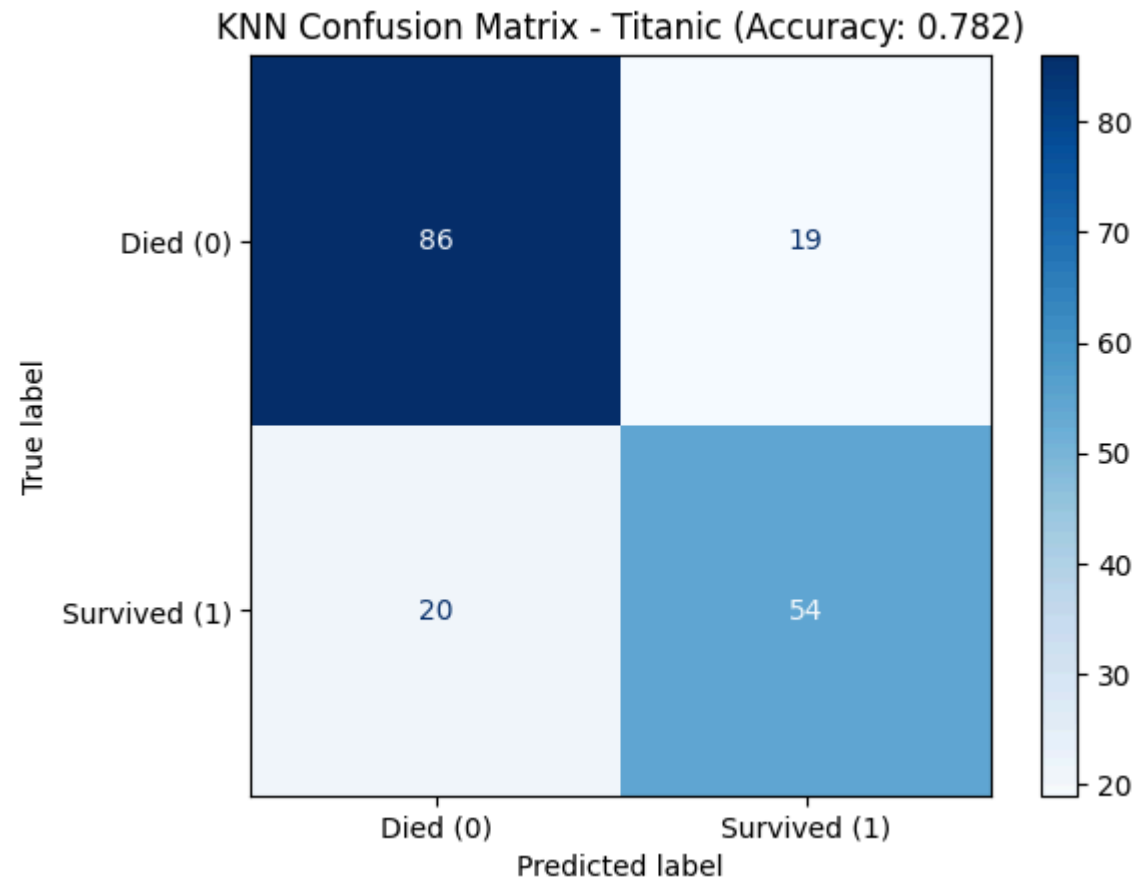
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Died (0)", "Survived (1)"])
disp.plot(cmap="Blues", values_format="d")
plt.title(f"KNN Confusion Matrix - Titanic (Accuracy: {acc:.3f})")
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plt.savefig("confusion_matrix_knn.png", dpi=150, bbox_inches="tight")  
plt.show()
```

KNN accuracy: 0.7821229050279329

Confusion Matrix:

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[[86 19]  
 [20 54]]
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In []: